
SOFTWARE REQUIREMENTS SPECIFICATION

for

Software Risk Tracker

Release 1.0

Version 1.0 approved

Prepared by Emily, Reddy, Logan, Nick, Alli, Brandon, Hannah, Josh, and Shelby

Contents

List of Figures	3
1 INTRODUCTION	5
1.1 Purpose	5
1.2 Document Conventions	5
1.3 Acronyms	6
1.4 Project Scope	6
2 OVERALL DESCRIPTION	7
2.1 Product Perspective	7
2.2 Product Features	8
2.3 Operating Environment	9
2.4 Design and Implementation Constraints	9
3 NONFUNCTIONAL REQUIREMENTS	10
3.1 Performance Requirements	10
3.2 Safety Requirements	11
3.3 Security Requirements	11
3.4 Software Quality Attributes	12
4 STANDARDS AND REFERENCES	13
4.1 Applicable Standards	13
4.2 References	15

List of Figures

2.1 Risk Management Matrix Example	7
--	---

Revision History

Date	Description	Revised by
10/21/25	Initial	Risk Team
01/14/26	Updated Requirements	Risk Team

1 INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) document will set the standards for software that will identify and track software project risks.

1.2 Document Conventions

This document is separated into 4 sub-sections, with each respective part then split further to thoroughly explain the process of development along with initial, intermediary with remedial steps, and final development stages. It describes an Introduction, Overall Description, Nonfunctional Requirements, as well as Standards and References. Each of the sections and sub-sections will follow the RFC-2119 recommended practices and standards engineering requirements. A Revision History, Table of Contents, and List of Figures is implemented for further clarity.

1.3 Acronyms

Acronym	Description
SRS	Software Requirements Specification
IEEE	Institute of Electrical and Electronics Engineers
ISO	International Organization for Standardization
IEC	International Electrotechnical Commission
SEI	Secure Coding Institute
WCAG	Web Content Accessibility Guideline

1.4 Project Scope

The scope of this project is to build a tool that aids software teams manage risks for their developed software. Risks will be identified, assessed, and tracked using this tool. Each risk will be qualified with a unique identifier, description, likelihood, and impact rating. The key feature of this tool will be the generation of risk matrices to visually represent the severity of risks. To ensure visibility and modification rights are given only to authorized users, access controls will be implemented. Furthermore, other design constraints and security/safety requirements will be within the scope of this project.

Outside the scope of this project will be AI-based features and third-party integrations.

2 OVERALL DESCRIPTION

2.1 Product Perspective

The software is designed to track and display risks for specific teams that are designing software. Users can create projects and store all risk data into our software. Then, based on the information given it will be stored and organized into a risk matrix similar to Figure 1. This is to simplify the view of the risks in order to manage risks.

5x5 Risk Matrix Example

Impact
How severe would the outcomes be if the risk occurred?

Probability
What is the probability the risk will happen?

	Insignificant 1	Minor 2	Significant 3	Major 4	Severe 5
5 Almost Certain	Medium 5	High 10	Very high 15	Extreme 20	Extreme 25
4 Likely	Medium 4	Medium 8	High 12	Very high 16	Extreme 20
3 Moderate	Low 3	Medium 6	Medium 9	High 12	Very high 15
2 Unlikely	Very low 2	Low 4	Medium 6	Medium 8	High 10
1 Rare	Very low 1	Very low 2	Low 3	Medium 4	Medium 5

SafetyCulture

Figure 2.1: Risk Management Matrix Example

2.2 Product Features

The user can create a new project, input risk data into the project, and the risk data is stored and organized into a visual matrix. Users can create a new project that has two levels of access. Those two levels of access will be admin and user. Admin has the ability to access any project on the software, add people to the project, and kick people off projects. Regular users will be able to make projects and edit information of projects they own or have access to. The Software will require the users authentication. The software will require the data on risk such as the following: the risk identifier, description of the risk, likelihood, and impact. The user will provide dimensions of the risk matrix upon creating the project. Once the user has input the matrix size and risk data then the software will store it and output it with a CSV file. The software will then display a table of the risk matrix that is color coded green to red based off likelihood and impact. There must be a user manual provided with the system upon delivery.

FUNC-1: The system shall allow users to create a new project by defining a project name.

FUNC-2: The system shall require user authentication (username and password) before granting access to project data.

FUNC-3: The system shall allow users to input risk data including a unique Risk Identifier, Description, Likelihood, and Impact.

FUNC-4: The system shall allow the user to define the dimensions (number of rows and columns) of the risk matrix during project creation.

FUNC-5: The system shall store all input data and provide a feature to export the project data as a CSV file.

FUNC-6: The system shall display a visual risk matrix table, dynamically color-coded from green to red based on the calculated risk level (Likelihood

× Impact).

FUNC-7: Regular users can access and edit the project information.

FUNC-8: Admin users can delete projects, add users to projects, remove users from projects, edit projects, and create and delete users.

FUNC-9: The software will comply with WCAG Guidelines.

2.3 Operating Environment

The software's intended users are software developers. It will run on the CS Server and will be accessed through a web browser. The software will comply with WCAG Guidelines. [5]

2.4 Design and Implementation Constraints

This application must be developed using the C++ programming language and implemented with the Webtoolkit (Witty) framework. Additionally, it must be capable of running on the computer science servers and must support multiple users.

3 NONFUNCTIONAL REQUIREMENTS

3.1 Performance Requirements

PERF-1: The system shall support up to 20 concurrent users while maintaining an average response time for all user actions under 2 seconds.

PERF-2: The response time to create, edit, or delete a risk shall be less than 2 seconds.

PERF-3: The response time to save user input for a risk (e.g., identifier, description, likelihood) shall be less than 1 second.

PERF-4: The response time for loading any new page (e.g., dashboard after login, home screen) shall be less than 5 seconds.

PERF-5: The response time to generate a complete risk matrix shall be less than 4 seconds.

PERF-6: The response time to update the risk matrix after a risk is created, edited, or deleted shall be less than 2 seconds.

PERF-7: The time it takes the system to export risk data to CSV shall be less than 2 seconds.

3.2 Safety Requirements

For the purposes of this system, safety is defined as follows: The system shall not pose any threat to human safety and shall operate in a manner that prevents unauthorized access and unintended loss of data.

SAFE-1: The system shall require a user to confirm any deletion action.

SAFE-2: The system shall automatically log out any user session that has been inactive for 15 minutes.

3.3 Security Requirements

SECU-1: The system shall require successful user authentication before granting access to any application functionality.

SECU-2: User data shall be stored in an SQLite database.

SECU-3: Will keep a transaction log of changes.

SECU-4: Passwords shall be stored with a secure hashing algorithm.

SECU-5: User activity shall be recorded in a log for the following actions: login, action performed (create, delete, edit), and CSV file exported.

SECU-6: If a user initiates a password reset, the system shall send a time-limited, single-use code to the user's registered email address to facilitate the reset.

3.4 Software Quality Attributes

QUAL-1: The system shall follow the Mozilla coding style.

QUAL-2: The system shall follow the SEI secure coding style.

QUAL-3: The system shall meet the accessibility standards provided by WCAG 2.1. [5].

4 STANDARDS AND REFERENCES

4.1 Applicable Standards

NIST - National Institute of Standards and Technologies

This system shall adhere to the recommended standards set forth by NIST 800-63B-4 on authentication username and passwords. [3]

IEEE - Institute of Electrical and Electronics Engineers

This system shall adhere to the recommended standards set forth by the IEEE STD 830-199B. [2]

Mozilla C++ Code Style Guide

This system shall adhere to the code style set forth by Mozilla.

SEI - Secure Engineering Institute

The system shall adhere to the secure coding style set forth by SEI. [4]

WCAG 2.1 - Web Content Accessibility Guidelines

This system shall meet the accessibility standards set forth by WCAG 2.1 [5,

1].

4.2 References

- [1] ACCESS, L. Digital accessibility solutions and services. <https://www.levelaccess.com/homepage-ppc/>, 2025. Accessed: 2025-10-21.
- [2] OF ELECTRICAL, I., AND (IEEE), E. E. IEEE official website. <https://www.IEEE.org/>, 2025. Accessed: 2025-10-21.
- [3] OF STANDARDS, N. I., AND (NIST), T. National institute of standards and technology official website. <https://pages.nist.gov/800-63-4/sp800-63b/authenticators/>, 2025. Accessed: 2026-01-14.
- [4] SOFTWARE ENGINEERING INSTITUTE, C. M. U. Sei cert c++ coding standard – rule 09: Object oriented programming (oop). <https://wiki.sei.cmu.edu/confluence/pages/viewpage.action?pageId=88046338>, 2025. Accessed: 2025-10-21.
- [5] (W3C), W. W. W. C. Web content accessibility guidelines (wcag) 2.1. <https://www.w3.org/TR/WCAG21/>, 2018. Accessed: 2025-10-21.